

# XAVIER MARCHENA

## QA Lead | Senior Test Automation Engineer | Full Stack Developer

San José, Costa Rica



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### PROFESSIONAL SUMMARY

Versatile technology professional with 9+ years of experience spanning QA leadership, test automation, and full-stack development. Currently serving as QA Lead overseeing quality assurance for complex cloud migrations and event-driven architectures. Proven track record of architecting robust testing frameworks across multiple languages (C#, TypeScript, Python, Ruby, Java) while maintaining strong full-stack development capabilities with modern frontend (Angular, React, Next.js) and backend (Node.js, .NET, Flask, Python) technologies. ISTQB-certified with expertise in leading distributed teams, implementing CI/CD practices, and delivering scalable solutions from concept to production. Unique combination of quality engineering leadership and hands-on development experience enables comprehensive understanding of software delivery lifecycle.

### EDUCATION

#### Bachelor's Degree in Computer Systems Engineering - 2025

Universidad Fidélitas, San José, Costa Rica

### CERTIFICATIONS

- ✓ ISTQB Certified Tester, Advanced Level Test Automation Engineer (October 2022)
- ✓ ISTQB Certified Tester, Foundation Level (May 2022)

### TECHNICAL SKILLS

|                           |                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------|
| Programming Languages     | JavaScript, TypeScript, Python, Java, Ruby, PHP, C#                                                              |
| Test Automation           | Selenium, Appium, Jest, Playwright, Cypress, Postman, Artillery, JUnit, TestNG, xUnit.net                        |
| Frontend Development      | Angular, Ionic, Next.js, React, HTML5, CSS3, Responsive Design                                                   |
| Backend Development       | Node.js, .NET Core/Framework, Flask, Express.js, RESTful APIs, GraphQL, SocketIO, Redis, Celery                  |
| Event-Driven Architecture | Kafka, AWS SQS, Message Queues, Event Processors, Pub/Sub Patterns                                               |
| Cloud & DevOps            | AWS (SQS, CloudWatch, S3), Azure (B2C, DevOps), Docker, Jenkins, GitHub Actions, Terraform, CI/CD Pipelines      |
| Version Control           | GitHub, GitLab, Bitbucket, Git Workflow, Code Review Practices                                                   |
| Databases                 | MySQL, MongoDB, Microsoft SQL Server, Oracle SQL, Database Design                                                |
| Development Tools         | Visual Studio Code, GitHub Copilot, Claude Code, Fiddler, Postman, JMeter, Datadog                               |
| Methodologies             | Agile/Scrum, Test-Driven Development (TDD), Behavior-Driven Development (BDD), CI/CD, Microservices Architecture |

## PROFESSIONAL EXPERIENCE

### Smash | QA Lead

April 2025 – Present | Full Time

**Environment:** C# .NET, xUnit.net, GitHub, AWS, Visual Studio Code, Claude Code, Terraform, Azure DevOps, Playwright, Cypress, NuGet, JMeter, Datadog, SQS, Kafka

Leading QA initiatives for migration from monolithic .NET architecture to cloud-based, event-driven microservices. Responsible for quality assurance across the core Documents team, managing complex testing strategies for distributed systems including multiple APIs (Metadata, Content, Accounts) and event-driven workflows utilizing SQS and Kafka streams.

- ▶ Lead comprehensive QA strategy across the platform evolution team, overseeing the entire testing pyramid including Unit, Contract, Integration, E2E, Synthetic, and Performance testing for event-driven architecture.
- ▶ Architect and implement testing strategies for distributed systems with event streaming, ensuring data integrity and message flow validation across SQS and Kafka streams consumed by downstream teams (OCR, Search, Navigation).
- ▶ Coordinate testing efforts for multiple API domains (Metadata, Content, Accounts) and event processors/publishers, ensuring seamless integration across microservices architecture.
- ▶ Partner with engineering managers and lead developers to continuously refine test strategies aligned with cloud migration initiatives, architectural changes, and business priorities.
- ▶ Establish and maintain QA best practices for event-driven systems, ensuring consistent quality standards across distributed teams handling asynchronous workflows.
- ▶ Mentor QA engineers on event-driven testing patterns, message queue validation, and distributed system quality assurance, facilitating knowledge sharing sessions to drive team growth.
- ▶ Drive automation coverage initiatives for both synchronous API testing and asynchronous event processing, reducing feedback cycles and improving release velocity for cloud-native applications.
- ▶ Evaluate and integrate emerging tools and methodologies for testing event-driven architectures, enhancing test robustness, scalability, and maintainability in cloud environments.

### Smash | Senior Quality Assurance Automation Engineer

July 2022 – April 2025 | Full Time

#### Modernization Team (Aug 2022 – July 2024)

**Environment:** C# .NET, xUnit.net, GitHub, Azure B2C, Visual Studio Code, GitHub Copilot, Terraform, Azure DevOps, Selenium, Cypress, NuGet

- ▶ Architected and developed three distinct test automation frameworks supporting the monolith-to-microservices modernization: E2E framework (C#, Selenium, FluentAssertions), Integration/API framework (xUnit, C#), and Frontend framework (Cypress for React).
- ▶ Designed and implemented a comprehensive Testing SDK with reusable client methods for API testing, packaged as a NuGet library for seamless integration across testing environments.
- ▶ Collaborated with distributed remote teams to strategically plan and execute the refactoring initiative, ensuring quality gates throughout the migration.
- ▶ Leveraged Azure DevOps for test case management, progress tracking, and defect reporting, maintaining traceability during the architectural transition.

#### Organizers Team (Jul 2022 – Aug 2023)

**Environment:** C# .NET, xUnit.net, Microsoft SQL Server, Azure DevOps, Fiddler, PowerShell, TestStack.White, FlaUI, Microsoft.Interop

- ▶ Conducted comprehensive manual testing while collaborating closely with development teams to ensure timely test case creation aligned with software requirements.
- ▶ Executed test plans during sprints and optimized the company's existing test automation solution for improved efficiency and maintainability.
- ▶ Developed and maintained detailed test plans, test cases, and test suites within Azure DevOps, ensuring thorough test coverage for each sprint.
- ▶ Utilized Fiddler for API traffic monitoring and analysis, simulating error responses to validate application error-handling capabilities.
- ▶ Implemented desktop automation using TestStack.White, FlaUI, and Microsoft.Interop for comprehensive UI testing coverage.

### RatherLabs | Performance Test Engineer

November 2022 – February 2023 | Part-Time Contractor

**Environment:** Artillery, Playwright, Node.js, Docker, Postman, AWS CloudWatch

- ▶ Performed comprehensive performance testing for a game in development, leveraging Artillery and Playwright to simulate server load and automate client interactions.
- ▶ Containerized automation code with Docker to ensure consistent, isolated test environments and enable multi-environment testing without dependency conflicts.

- ▶ Collaborated with the development team in Agile ceremonies (daily standups, sprint planning, code reviews) to align test coverage with requirements and catch issues early.
- ▶ Created mock servers using Postman to reduce AWS resource dependency, resulting in cost savings and improved testing efficiency.
- ▶ Monitored and analyzed test results via AWS CloudWatch and Artillery logs, identifying performance bottlenecks and collaborating with developers on optimization strategies.

**Proximity (10Pearls Company) | Senior QA Automation Engineer**

April 2022 – July 2022 | Full Time

**Environment:** Playwright, Artillery, TypeScript, Node.js, Postman

- ▶ Developed proof-of-concept test automation solution for web GUI and API testing using Artillery, Playwright, and TypeScript.
- ▶ Designed and implemented automated test cases for both browser and API components by analyzing software requirements and developing comprehensive validation tests.
- ▶ Configured testing framework for repeatable and consistent test execution, resolving issues and optimizing test reliability.
- ▶ Specialized in API testing using Postman, designing test cases for multiple endpoints and methods while verifying response accuracy.
- ▶ Leveraged Postman's advanced features (test scripts, collections, environments) to automate repetitive tasks and improve testing efficiency.
- ▶ Partnered with development team to identify and resolve API-related issues including performance, security, and functionality concerns.

**Proximity (10Pearls Company) | Mid-Level QA Automation Engineer**

June 2021 – April 2022 | Full Time

**Environment:** Appium, Ruby, Docker, Jenkins, Jest, Selenium, BrowserStack

- ▶ Architected and developed iOS app test automation framework using Appium, Ruby, Docker, and Jenkins, implementing reusable code libraries and best practices.
- ▶ Automated iOS app testing across real devices and emulators, including Appium server setup and Ruby script development.
- ▶ Containerized test automation code with Docker to streamline cross-environment execution and accelerate team member onboarding.
- ▶ Integrated test framework with Jenkins for automated continuous testing, enabling timely quality feedback and supporting CI/CD practices.
- ▶ Created automated tests for React application using Jest, Selenium, and BrowserStack, covering diverse scenarios and user flows across multiple browsers and devices.
- ▶ Collaborated with development team using JIRA and GitHub for issue tracking, management, and code reviews.

**OMNi Costa Rica | Quality Assurance Test Engineer**

August 2020 – June 2021 | Full Time

**Environment:** iOS, React, Android, MySQL, AWS, Jenkins, Bitbucket, Postman, Jest, Docker, DataDog

- ▶ Conducted manual testing for web and mobile applications, ensuring functionality and quality aligned with requirements.
- ▶ Performed extensive API testing using Postman, validating API calls and responses for system reliability and consistency.
- ▶ Reviewed and executed Jest-based automated test suite, ensuring compliance with specifications and requirements.
- ▶ Developed RPA solutions to automate repetitive tasks, increasing testing process efficiency.
- ▶ Participated in integration, regression, and production testing to ensure software quality and expected performance.
- ▶ Analyzed system and business requirement documents, developing test scenarios and creating detailed test cases in Xray.
- ▶ Managed defects in JIRA with proper tracking and resolution workflows in Agile environment.

**RGT Global | Full Stack Developer**

January 2020 – August 2020 | Full Time

**Environment:** Python, Angular, Ionic, Flask, MySQL, MongoDB, Redis, Celery, Docker, SocketIO, NumPy, Pandas

- ▶ Architected and developed scalable microservices-based Reports API using Python, Flask, NumPy, and Pandas in Docker containers, implementing data aggregation and analysis features for business intelligence reporting.
- ▶ Built cross-platform mobile application with Ionic and Angular, implementing responsive UI/UX design patterns and ensuring consistent experience across iOS, Android, and web platforms.
- ▶ Designed and implemented secure payment processing system integrated with multiple payment gateways (Stripe, PayPal), incorporating PCI compliance standards and transaction validation logic using Python and Angular.
- ▶ Developed full-stack real-time betting game proof of concept featuring WebSocket communication (SocketIO), demonstrating proficiency in building interactive applications with Angular frontend, Python backend, and MongoDB for data persistence.
- ▶ Implemented asynchronous task processing using Celery with Redis message broker for handling background jobs, improving application responsiveness and scalability for high-volume operations.
- ▶ Performed advanced data analysis and ETL operations using Python, NumPy, and Pandas, creating automated reporting pipelines that provided actionable business insights to stakeholders.

- ▶ Collaborated with product team in Agile environment, participating in sprint planning, daily standups, and retrospectives while delivering iterative improvements based on user feedback and business requirements.
- ▶ Containerized all applications using Docker, creating reproducible development and production environments that streamlined deployment processes and reduced environment-related issues.

**QvoTech | QA Automation Engineer**

August 2019 – January 2020 | Full Time

**Environment:** Selenium, MySQL, Python, Unittest, Linux, Postman, JIRA

- ▶ Identified and documented detailed requirements for testing system functionality and needs.
- ▶ Developed automated tests using Python, Selenium, and Unittest framework based on user stories and acceptance criteria.
- ▶ Performed comprehensive testing including API, integration, user acceptance, regression, and production testing.
- ▶ Created mock servers using Postman to simulate API responses and reduce external system dependencies.
- ▶ Developed and maintained testing documentation including test cases, test plans, and test scripts.
- ▶ Tracked defects in JIRA and collaborated with development team for timely issue resolution.

**Universidad de Costa Rica | Software Developer**

January 2018 – January 2019 | Full-Time Contractor

**Environment:** Java, PHP, WordPress, Docker, LibreOffice SDK, Shell Script, Batch Script, Git

- ▶ Led research and development initiatives focused on Free and Open Source Software (FOSS) solutions, evaluating technologies and implementing systems to benefit university operations and reduce software licensing costs.
- ▶ Developed custom Java extension for LibreOffice using the LibreOffice SDK, implementing automation features and custom functionality that improved document processing efficiency for university administrative staff.
- ▶ Architected and maintained WordPress-based institutional website, implementing custom themes and plugins while establishing Docker-based development environment to ensure consistency across development, staging, and production.
- ▶ Designed and developed cross-platform software installer using Java, implementing platform detection logic and automated deployment scripts (Shell Script for Linux/macOS, Batch Script for Windows) to simplify software distribution across heterogeneous university IT infrastructure.
- ▶ Created comprehensive technical documentation and training materials, delivering hands-on training sessions to university staff to facilitate adoption of newly developed tools and improve technical literacy.
- ▶ Collaborated with university departments to identify workflow inefficiencies and pain points, developing custom automation solutions that reduced manual processes and improved operational efficiency.
- ▶ Implemented version control workflows using Git, conducting code reviews and establishing development best practices for the university's software development initiatives.
- ▶ Applied Agile methodologies for project management, participating in sprint planning and delivering incremental improvements based on stakeholder feedback and evolving requirements.

**Tek Experts | Technical Support Engineer**

August 2016 – December 2016 | Full Time

**Environment:** Windows Server, Linux, Microsoft SQL, Oracle SQL, HP ALM/Quality Center

- ▶ Installed and configured HP ALM/Quality Center on Windows and Linux environments, ensuring full functionality.
- ▶ Provided technical support for database installation, upgrades, migrations, and restoration (Oracle and SQL Server).
- ▶ Delivered telephone and email support to users and administrators, providing troubleshooting guidance.
- ▶ Resolved application errors by researching knowledge bases and analyzing client logs to identify root causes.
- ▶ Developed automation scripts for repetitive tasks including database backups and restores.
- ▶ Created and maintained comprehensive documentation for installation, configuration, and troubleshooting processes.